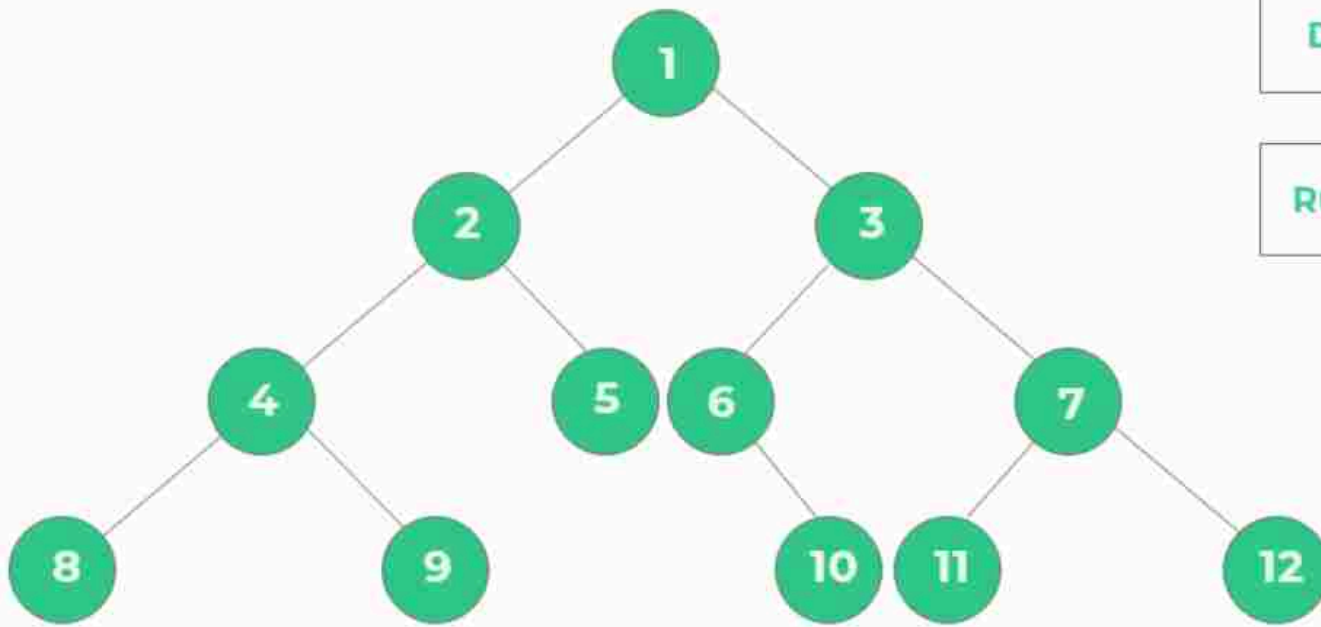




Direction : Anti Clock

Rule : CLR (Center-Left-Right)

PreOrder:

1 2 4 8 9 5 3 6 10 7 11 12

Constraints:

0<size <100

0<data<1000

Input format:

First line indicates the size of the tree

Second line indicates the elements of the tree

Output Format:

single line indicates the pre-order list of the given value

Logical Test Cases

Test Case 1

INPUT (STDIN)

7
200 120 80 160 280 240 320

EXPECTED OUTPUT

200 120 80 160 280 240 320

Test Case 2

INPUT (STDIN)

9
34 67 10 90 410 810 40 20 60

EXPECTED OUTPUT

34 10 20 67 40 60 90 410 810

Mandatory Test Cases

Test Case 1

KEYWORD

void preorder(struct node* root)

Test Case 2

KEYWORD

struct node* newNode(int item)

Test Case 3

KEYWORD

struct node

Test Case 4

KEYWORD

struct node *left,*right;

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

280

Test Case 3

NLOC

50

Tree11.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *left,*right;
7 };
8
9 struct node* newNode(int item) {
10     struct node* temp = (struct node*)malloc(sizeof(struct node));
11     temp->data = item;
12     temp->left = NULL;
13     temp->right = NULL;
14     return temp;
15 }
16
17 struct node* insert(struct node* node, int data) {
18     if (node == NULL) return newNode(data);
19     if (data < node->data)
20         node->left = insert(node->left, data);
21     else if (data > node->data)
22         node->right = insert(node->right, data);
23     return node;
24 }
```

```
26 void preorder(struct node* root) {  
27     if (root != NULL) {  
28         printf("%d ", root->data);  
29         preorder(root->left);  
30         preorder(root->right);  
31     }  
32 }  
33  
34 int main() {  
35     int n, val;  
36     if (scanf("%d", &n) != 1) return 0;
```

```
37  struct node* root = NULL;
38  for (int i = 0; i < n; i++) {
39      if (scanf("%d", &val) == 1) {
40          root = insert(root, val);
41      }
42  }
43  preorder(root);
44  printf("\n");
45  return 0;
46}
47
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
7
```

```
200 120 80 160 280 240 320
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree11.exe
```

```
200 120 80 160 280 240 320
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
9
```

```
34 67 10 90 410 810 40 20 60
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree11.exe
```

```
34 10 20 67 40 60 90 410 810
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```